ESE 2150 Lab 2

Wheatstone Bridge Characterization and LED Resistance

Ritvik Ellendula Sudhir Yelamarthy

Fall 2026

A. Wheatstone Bridge Characterization

1. Measurement Procedure and Observations

First, we built the Wheatstone bridge circuit. We used two benchtop multimeters as our voltmeters, connecting their positive and negative leads to the nodes shown in the circuit diagram. We turned on the power supply and adjusted the potentiometer until the Channel 1 voltmeter reading was as close to 0 V as possible with both buttons unpressed. We then turned off the supply, pressed both buttons, turned the supply back on, and recorded the voltage on Channel 2 without changing the potentiometer setting.

We repeated this procedure for $R_x = 2.2\,\mathrm{k\Omega}$, $4.7\,\mathrm{k\Omega}$, and $10\,\mathrm{k\Omega}$, first with a 2 V supply and then with a 3 V supply. We found that the Channel 2 voltage decreased as R_x increased. This makes sense because the balanced bridge satisfies

$$\frac{4.7\,\mathrm{k\Omega}}{R_2} = \frac{R_x}{4.7\,\mathrm{k\Omega}}.$$

As R_x increases, R_2 must decrease to maintain this equality. When both buttons are pressed, Channel 2 measures the voltage across R_2, so a smaller R_2 produces a smaller measured voltage. We observed the same trend at both supply voltages.

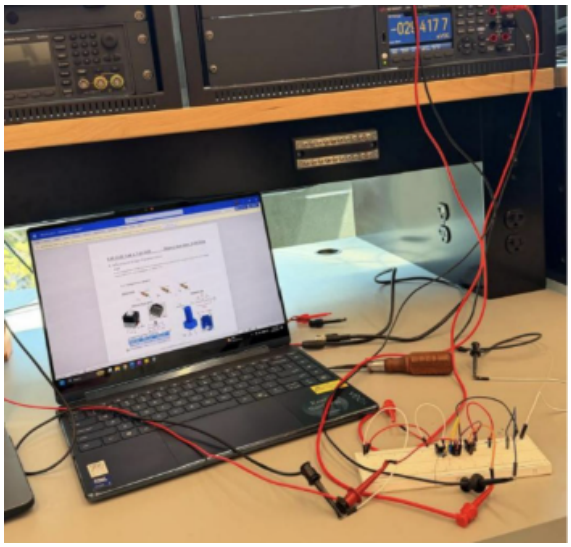


Figure 1: Wheatstone bridge setup for Part A.

2. Measurement Results

V_s (V)	Actual R_x (Ω)	Ch1 (mV)	Ch2 (V)	Calculated R_2 (Ω)	Calculated R_x (Ω)	Difference (Ω)
2	2200	-29.4	1.990	9950.00	2220.10	-20.10
2	4700	-0.210	0.979	4895.00	4512.77	187.23
2	10000	2.8	0.459	2295.00	9625.27	374.73
3	2200	-44	2.990	9966.67	2216.39	-16.39
3	4700	-5	1.460	4866.67	4539.04	160.96
3	10000	-2.3	0.682	2273.33	9717.01	282.99

Table 1: Measurement and calculation results for Part A. Ch1 was measured with both switches unpressed; Ch2 was measured with both switches pressed.

3. Calculations

With both switches pressed, the potentiometer forms a voltage divider. Since $R_1 + R_2 = 10\,\mathrm{k}\Omega$,

$$V_{\mathrm{Ch2}} = \frac{R_2}{R_1 + R_2} V_s = \frac{R_2}{10\,000} V_s.$$

Solving for R_2,

$$\boxed{R_2 = \frac{V_{\mathrm{Ch2}}}{V_s}(10\,000\,\Omega)}.$$

At balance, the two middle nodes have the same voltage, giving

$$\frac{R_x}{4700} = \frac{4700}{R_2}.$$

Therefore,

$$\boxed{R_x = \frac{(4700\,\Omega)^2}{R_2}}.$$

$V_s = 2\,\mathrm{V}$

For the nominal $2.2\,\mathrm{k}\Omega$ resistor,

$$R_2 = \frac{1.990}{2}(10\,000) = 9950\,\Omega,$$
$$R_x = \frac{4700^2}{9950} = 2220.10\,\Omega.$$

For the nominal $4.7\,\mathrm{k}\Omega$ resistor,

$$R_2 = \frac{0.979}{2}(10\,000) = 4895\,\Omega,$$
$$R_x = \frac{4700^2}{4895} = 4512.77\,\Omega.$$

For the nominal $10\,\mathrm{k}\Omega$ resistor,

$$R_2 = \frac{0.459}{2}(10\,000) = 2295\,\Omega,$$
$$R_x = \frac{4700^2}{2295} = 9625.27\,\Omega.$$

$V_s = 3\,\mathrm{V}$

For the nominal $2.2\,\mathrm{k}\Omega$ resistor,

$$R_2 = \frac{2.990}{3}(10\,000) = 9966.67\,\Omega,$$
$$R_x = \frac{4700^2}{9966.67} \approx 2216.39\,\Omega.$$

For the nominal $4.7\,\mathrm{k}\Omega$ resistor,

$$R_2 = \frac{1.460}{3}(10\,000) = 4866.67\,\Omega,$$
$$R_x = \frac{4700^2}{4866.67} \approx 4539.04\,\Omega.$$

For the nominal $10\,\mathrm{k}\Omega$ resistor,

$$R_2 = \frac{0.682}{3}(10\,000) = 2273.33\,\Omega,$$
$$R_x = \frac{4700^2}{2273.33} \approx 9717.01\,\Omega.$$

4. Discussion

We see a slight difference between the calculated and nominal resistances because we did not obtain an exact voltage difference of 0 V between the two middle nodes. We could not perfectly adjust the potentiometer to make the ratios match. Resistor tolerances, wire and breadboard contact resistances, and measurement uncertainty also contribute to the differences.

The resistance of R_x does not depend on the supply voltage under the conditions of this experiment. We obtained values close to the nominal resistances at both 2 V and 3 V. The small differences between the two sets of calculated values reflect the experimental limitations rather than a strong voltage dependence of the resistors.

B. LED Resistance Measurement Using a Wheatstone Bridge

1. Measurement Procedure and Observations

First, we built the same Wheatstone bridge circuit as in Part A, but replaced R_x with a red LED in the correct orientation. We used the two benchtop multimeters as Channels 1 and 2 and connected them as shown in the circuit diagram.

For each supply voltage, we left both buttons unpressed and adjusted the potentiometer until the Channel 1 voltage was close to 0 V. We then turned off the power, pressed both buttons, and turned the power back on. We recorded the Channel 2 voltage, which measures the voltage across R_2, and used it to calculate R_2. Using the Wheatstone bridge balance equation, we calculated the effective resistance of the LED at that operating condition. We then turned off the supply, released the buttons, changed the supply voltage, and repeated the procedure.

We found that the effective resistance of the LED decreased as the supply voltage increased. This decrease was not proportional to the change in supply voltage. The resistance at 1.6 V was especially high, and the Channel 2 voltage increased as the supply voltage increased. These observations suggest that the LED is nonlinear. The LED became visibly illuminated at around a 3 V supply.

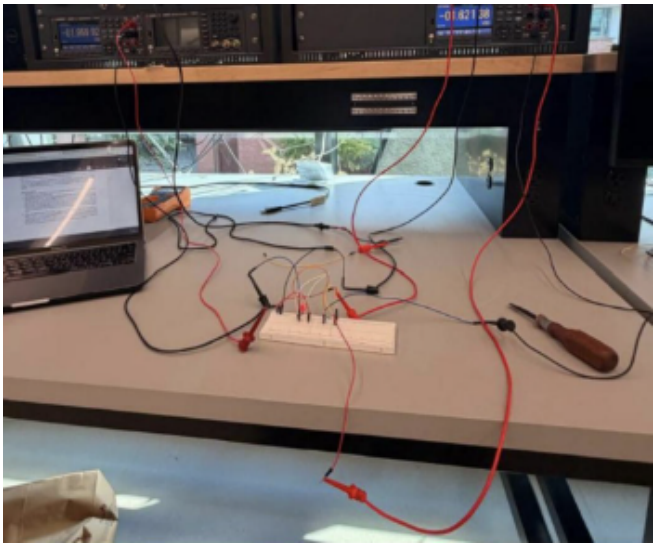


Figure 2: LED Wheatstone bridge setup for Part B.

2. Measurement Results

Supply voltage (V)	Ch1 (mV)	Ch2 (V)	R_2 (Ω)	R_{LED} (Ω)
1.6	1.5	0.0460	287.50	76834.78
1.7	-2.0	0.0887	521.76	42337.09
1.8	-2.7	0.1380	766.67	28813.04
2.0	1.0	0.2660	1330.00	16609.02
2.2	-5.6	0.4080	1854.55	11911.27
2.5	-6.7	0.6740	2696.00	8193.62
3.0	7.0	1.2500	4166.67	5301.60
3.5	-4.0	1.9500	5571.43	3964.87
4.0	-7.0	2.7900	6975.00	3167.03
4.5	0.8	3.8000	8444.44	2615.92
5.0	-6.7	4.9000	9800.00	2254.08

Table 2: Measurement and calculation results for Part B. Ch1 was measured with both switches unpressed; Ch2 was measured with both switches pressed.

3. Calculations

The same voltage-divider relationship gives

$$R_2 = \frac{V_{\text{Ch2}}}{V_s}(10\,000\,\Omega).$$

At bridge balance, the effective LED resistance is

$$R_{\text{LED}} = \frac{(4700\,\Omega)^2}{R_2}.$$

For each supply voltage, the calculations are as follows. Intermediate values are displayed rounded; final results use the unrounded values.

$$V_s = 1.6\,\text{V}: \quad R_2 = \frac{0.046}{1.6}(10\,000) = 287.50\,\Omega,$$
$$R_{\text{LED}} = \frac{4700^2}{287.50} = 76834.78\,\Omega.$$

$$V_s = 1.7\,\text{V}: \quad R_2 = \frac{0.0887}{1.7}(10\,000) \approx 521.76\,\Omega,$$
$$R_{\text{LED}} = \frac{4700^2}{(0.0887/1.7)(10\,000)} \approx 42337.09\,\Omega.$$

$$V_s = 1.8\,\text{V}: \quad R_2 = \frac{0.138}{1.8}(10\,000) \approx 766.67\,\Omega,$$
$$R_{\text{LED}} = \frac{4700^2}{(0.138/1.8)(10\,000)} \approx 28813.04\,\Omega.$$

$$V_s = 2.0\,\mathrm{V}: \quad R_2 = \frac{0.266}{2}(10\,000) = 1330\,\Omega,$$
$$R_{\mathrm{LED}} = \frac{4700^2}{1330} \approx 16609.02\,\Omega.$$

$$V_s = 2.2\,\mathrm{V}: \quad R_2 = \frac{0.408}{2.2}(10\,000) \approx 1854.55\,\Omega,$$
$$R_{\mathrm{LED}} = \frac{4700^2}{(0.408/2.2)(10\,000)} \approx 11911.27\,\Omega.$$

$$V_s = 2.5\,\mathrm{V}: \quad R_2 = \frac{0.674}{2.5}(10\,000) = 2696\,\Omega,$$
$$R_{\mathrm{LED}} = \frac{4700^2}{2696} \approx 8193.62\,\Omega.$$

$$V_s = 3.0\,\mathrm{V}: \quad R_2 = \frac{1.25}{3}(10\,000) \approx 4166.67\,\Omega,$$
$$R_{\mathrm{LED}} = \frac{4700^2}{(1.25/3)(10\,000)} = 5301.60\,\Omega.$$

$$V_s = 3.5\,\mathrm{V}: \quad R_2 = \frac{1.95}{3.5}(10\,000) \approx 5571.43\,\Omega,$$
$$R_{\mathrm{LED}} = \frac{4700^2}{(1.95/3.5)(10\,000)} \approx 3964.87\,\Omega.$$

$$V_s = 4.0\,\mathrm{V}: \quad R_2 = \frac{2.79}{4}(10\,000) = 6975\,\Omega,$$
$$R_{\mathrm{LED}} = \frac{4700^2}{6975} \approx 3167.03\,\Omega.$$

$$V_s = 4.5\,\mathrm{V}: \quad R_2 = \frac{3.8}{4.5}(10\,000) \approx 8444.44\,\Omega,$$
$$R_{\mathrm{LED}} = \frac{4700^2}{(3.8/4.5)(10\,000)} \approx 2615.92\,\Omega.$$

$$V_s = 5.0\,\mathrm{V}: \quad R_2 = \frac{4.9}{5}(10\,000) = 9800\,\Omega,$$
$$R_{\mathrm{LED}} = \frac{4700^2}{9800} \approx 2254.08\,\Omega.$$

4. LED Resistance Versus Supply Voltage

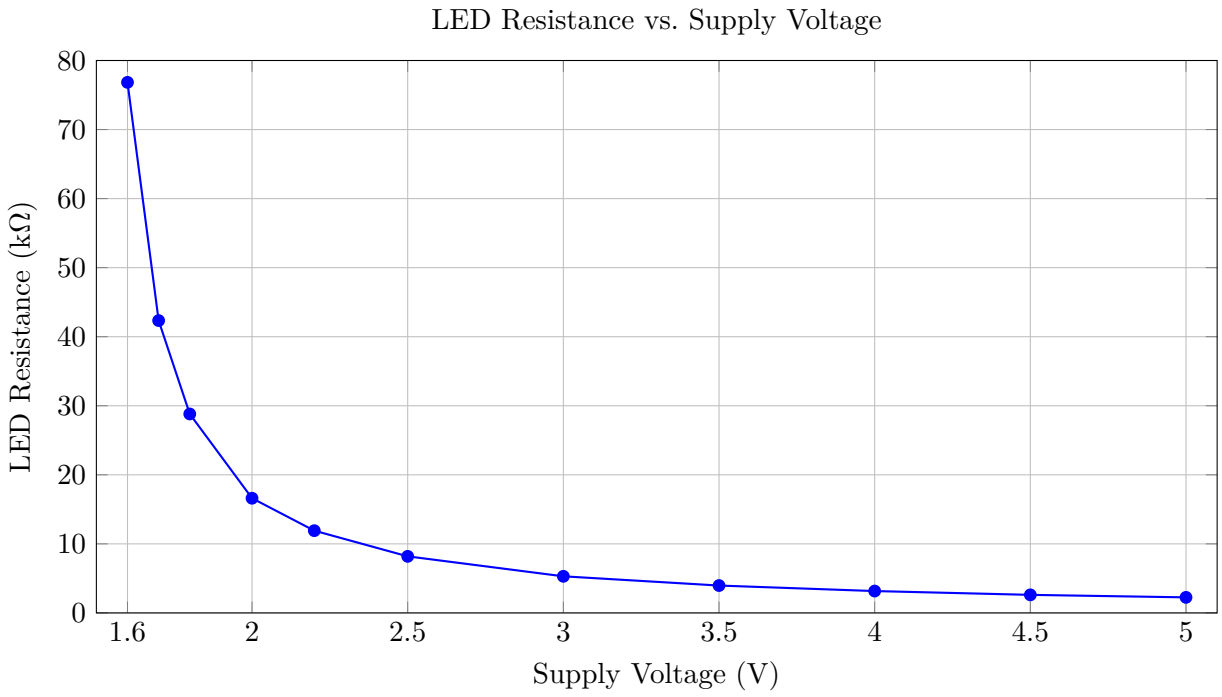


Figure 3: Effective LED resistance decreases nonlinearly as the supply voltage increases.

5. Is the Response Linear?

No, the response is not linear. The effective resistance of the LED changes by different amounts as the supply voltage changes. An equal increase in voltage does not produce an equal increase in current.

An LED is a diode, so its current depends nonlinearly on the voltage across its semiconductor p-n junction. As forward bias increases, the junction allows much more current to flow. Therefore, the ratio

$$R_{\text{LED}} = \frac{V_{\text{LED}}}{I_{\text{LED}}}$$

is not constant. This is different from an ideal resistor, which has a constant voltage-to-current ratio.

6. Emitted Light, Resistance, and Heat

The emitted light increased as the supply voltage increased because the greater forward bias allowed more current to flow through the LED. Over the measured range, the current increased more rapidly than the voltage across the LED, so its effective resistance decreased.

The resistance measured here represents the ratio of the LED voltage to its current at a particular operating point:

$$\boxed{R_{\text{LED}} = \frac{V_{\text{LED}}}{I_{\text{LED}}}}.$$

It describes the LED's behavior at that operating point; it is not a fixed component value. It is also

different from the differential resistance,

$$r_d = \frac{dV_{\text{LED}}}{dI_{\text{LED}}},$$

which describes how the voltage changes for a small change in current near that point.

The effective resistance does not directly measure generated heat. The electrical power delivered to the LED is

$$P_{\text{electrical}} = V_{\text{LED}} I_{\text{LED}}.$$

Some of this power becomes emitted light, while the remainder is primarily dissipated as heat:

$$P_{\text{heat}} \approx V_{\text{LED}} I_{\text{LED}} - P_{\text{light}}.$$

A decrease in effective resistance does not necessarily mean less heat is generated; the electrical power can increase as the LED current increases.

7. Could an Ohmmeter Replace the Wheatstone Bridge?

An ohmmeter could be used to measure the ordinary resistors in Part A, but it would not replace the Wheatstone bridge for the LED measurements in Part B.

An ohmmeter applies its own test current or voltage and determines resistance from the resulting voltage-to-current ratio. For an approximately linear resistor, this gives a useful resistance value because that ratio is nearly constant.

For an LED, the voltage-to-current ratio depends on the operating point. The ohmmeter would select the test condition rather than allow us to characterize the LED at the different supply voltages used in this experiment. Its test signal might also be too small to forward bias the LED sufficiently.

The Wheatstone bridge allows us to determine the LED's effective resistance at each operating condition established by the circuit. An ohmmeter reading at its own test condition would not provide the same set of measurements.